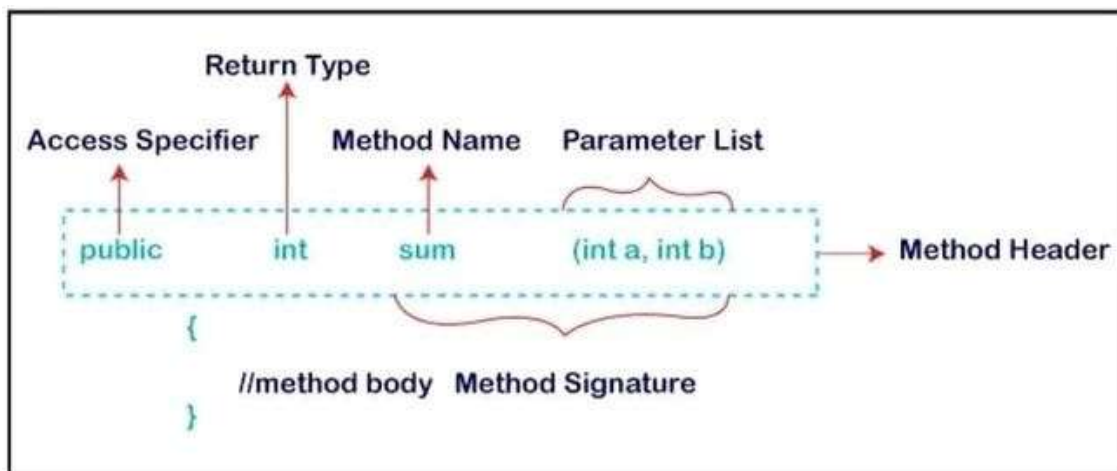


Parameterized Constructor



What is Parameterized Constructor ?

- A parameterized constructor is a special method within a class that accepts **one or more arguments (parameters)** when an object is created.



How it is Work :-

- **Declaration:** A parameterized constructor is declared with parameters, similar to a regular function.
- **Object Creation:** When you create an object of the class, the required arguments to the constructor.
- **Initialization:** The values of these arguments are then used to set the initial state of the object's attributes.

Implementation of Parameterized Constructor

Input :-

```
1 package constructor;
2
3 public class parameterized_constructor {
4
5     parameterized_constructor(int a)
6     {
7         System.out.println("Hello Pune");
8     }
9
10    parameterized_constructor(int a,int b,int c)
11    {
12        System.out.println("Multiplication is : " +(a*b*c));
13    }
14
15    parameterized_constructor(int a,int b,int c, int d)
16    {
17        System.out.println("Add is : " +(a+b+c+d));
18    }
19
20    public static void main(String[] args) {
21
22        parameterized_constructor p=new parameterized_constructor(2);;
23        parameterized_constructor p2=new parameterized_constructor(2,3,4);
24        parameterized_constructor p3=new parameterized_constructor(4,8,12,6);
25    }
26 }
```

Output :-

```
Hello Pune
Multiplication is : 24
Add is : 30
```

Copy Constructor



What is Copy Constructor :-

- A Copy Constructor is an **overloaded constructor**.
- When an object is passed to a function, a bitwise copy of that **object is made and given to the function**.
- If the object contains a **pointer to allocated memory**.
- The copy will point to the **memory as does the original object**.
-

Copy Constructor in Java

```
constructor 1{}
```

```
Constructor2(constructor 1)  
{}  
}
```



Purpose and Use :-

- Deep Copying
- Object Initialization
- Design Patterns

Implementation of Copy Constructor

Input :-

```
1 package constructor;
2 class Employee
3 {
4     int emp_id;
5     String emp_name;
6
7     // Parameterized constructor
8 Employee(int emp_id, String emp_name)
9 {
10    this.emp_id = emp_id;
11    this.emp_name = emp_name;
12    System.out.println("emp id : " + this.emp_id);
13    System.out.println("emp name : " + this.emp_name);
14 }
15 // Copy constructor
16 Employee(Employee e)
17 {
18    this.emp_id = e.emp_id;
19    this.emp_name = e.emp_name;
20    System.out.println("Copy of emp id : " + this.emp_id);
21    System.out.println("Copy of emp name : " + this.emp_name);
22 }
23 }
24
25 public class copy_constructor {
26     public static void main(String[] args)
27     {
28         Employee emp1 = new Employee(23, "Nikhil");
29         System.out.println();
30         Employee emp2 = new Employee(emp1);
31     }
32 }
```

Output :-

```
emp id : 23
emp name : Nikhil

Copy of emp id : 23
Copy of emp name : Nikhil
```